

PRESCRIPTIVE PROFIT ENGINE: THE ADVANCED PRICING FRAMEWORK

SEBASTIAN THURM
LEAD DATA, ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

STRATEGIC OBJECTIVE: BRIDGING THE GAP BETWEEN BEHAVIORAL INTELLIGENCE AND EXECUTIVE CONTROL

Executive Impact:

In traditional retail, pricing is often a reactive process based on competitor tracking or historical sales. This project implements the Advanced Pricing Framework, moving the business from "What happened?" to "What should we do now?" By utilizing an Advanced XGBoost Regressor on AWS SageMaker, this engine analyzes high-dimensional behavioral data—specifically Cart-to-Detail (CtD) and Buy-to-Detail (BtD) ratios—to identify "Pricing Walls" in real-time. This allows for surgical price nudges that a human analyst or a simple linear model would miss.

Problem Statement:

- The Scalability Wall: Manual pricing teams can only manage a fraction of a 10,000+ SKU catalog, leaving the "Long Tail" of inventory to rot or sell at suboptimal prices.
- The "Gut-Feeling" Trap: Decisions are often reactive (based on what sold yesterday) rather than proactive (based on who is clicking today but not buying).
- Static Logic in a Dynamic Market: Traditional "Cost-Plus" pricing ignores Behavioral Friction, failing to identify when a price is a "barrier to entry" versus when a product is simply out of season.
- Inventory Deadlock: Capital stays locked in "Zombie Stock" because liquidation happens too late, leading to emergency fire sales that destroy brand equity.

Enterprise-Grade MLOps Infrastructure

To achieve "Advanced" status, the model is supported by a modern, decoupled AWS architecture that ensures reliability and cost-efficiency:

- CloudFormation (IaC): The entire stack—from S3 buckets to IAM security policies—is defined as code, allowing for "one-click" deployment across different regions or business units.
- SageMaker Feature Store: Acts as the centralized registry for our behavioral features. By storing "Friction" and "Pressure" scores here, we ensure that the model always uses the most recent customer data for its predictions.
- Managed Training & S3 Data Lake: High-volume raw data is staged in S3 and fed into SageMaker Training Jobs. This allows us to train complex models on massive datasets without managing underlying server hardware.
- Serverless Inference: The model is deployed via AWS Serverless Inference. This is a massive economic advantage, as it provides a "pay-per-request" cost model—the business pays \$0 when the site is quiet, and the system automatically scales to handle thousands of requests during peak sales events.

The "CEO Steering Wheel": Strategic Fixed Points

Advanced ML models are often criticized for being "Black Boxes." To bridge the gap between AI and Executive Leadership, this engine introduces Strategic Fixed Points (The Margin, Balanced, and Revenue Toggles). Why these "Fixed Points" are necessary for a CEO:

- Intentional Direction: A CEO needs to align pricing with the current quarterly goal (e.g., "We need cash now to buy new spring stock"). The toggles allow leadership to lead the AI, not just follow it.
- Risk Mitigation: The Guardrail System (e.g., max -50%) ensures that even the most aggressive AI signal cannot accidentally destroy brand equity by pricing a premium item at \$1.
- Dynamic Response: By adjusting a single multiplier, the CEO can shift the entire catalog's behavior from "Profit Maximization" to "Liquidation" in seconds.

TABLE 1: STRATEGIC INTELLIGENCE (INFERENCE LOGIC)							
SCENARIO NAME	STOCK	PRESS	FRICT	ELAST	MARGIN %	BALANCED %	REVENUE %
Dead Stock (Liquidation)	14500	0.97	0.00	1.50	-1.9%	-2.9%	-9.6%
Warehouse Emergency	13000	0.87	0.00	1.80	-2.2%	-3.3%	-11.0%
Legacy Model Clearance	9500	0.63	0.13	2.10	-50.0%	-50.0%	-50.0%
End of Season Bulk	11000	0.73	0.07	1.90	-50.0%	-50.0%	-50.0%
Overstocked Classic	4000	0.27	0.00	1.10	-0.3%	-0.5%	-1.7%
Viral Flop (High Frict)	900	0.06	0.75	2.50	-24.3%	-36.4%	-50.0%
Pricing Wall (Hard)	1500	0.10	0.88	2.90	-41.0%	-50.0%	-50.0%
Checkout Abandoner	2200	0.15	0.44	2.20	-50.0%	-50.0%	-50.0%
Clickbait Item	5000	0.33	0.92	2.40	-50.0%	-50.0%	-50.0%

Key Performance Outcomes:

- Behavioral Funnel Optimization: By isolating the Friction Score (CtD vs. BtD), the model identifies "Pricing Walls" in real-time, allowing for surgical interventions that increase conversion rates without store-wide discounting.
- Automated Liquidity Acceleration: The system automatically triggers the Revenue Strategy as inventory approaches the 15000 unit "Pressure Ceiling," ensuring warehouse throughput stays fluid and cash is never stagnant. This value can be also adjusted for individual needs.
- Precision Margin Defense: Unlike traditional rule-based engines, this model defends MSRP for "Superstars" and "Hidden Gems," preventing 5% - 15% in "accidental discounting" on items with high organic demand.
- Operational Cost Reduction: Leveraging AWS Serverless Inference, the business eliminates the overhead of a 24/7 manual pricing desk, scaling decision-making to zero cost during idle periods.

The Bottom Line Summary:

This is not just a pricing tool; it is a Capital Efficiency Engine. By integrating SageMaker's advanced machine learning with CloudFormation-backed infrastructure, we have transformed pricing from a back-office administrative task into a front-line competitive advantage. The system doesn't just predict a price—it prescribes a business outcome. Whether the goal is to defend the bottom line (Margin) or clear the warehouse floor (Revenue), the Strategic Toggles give leadership the "Steering Wheel" to lead the AI. In a real-world deployment, this framework transitions a retailer from a Reactive state (losing money on slow movers) to a Prescriptive state (optimizing every cent of inventory value).